

DIRECTIONS: Give one scenario or example within the given system that demonstrates each OOP principle.

1. Film Production System (10 points)

Scenario: A film studio is developing a Film Production Management System that manages actors, directors, crew members, scenes, and filming schedules.

Principle	Your Example
Encapsulation	There is a class called Crew but the information and details such as scenes and filmingSchedules are hidden and put on private and not on public so that it can only be accessed by a method within that class.
Abstraction	You call a function action() and all of the crew members acted. There is no “hey you do an up-close shot” or “hey, cry on this scene”. With just the function action(), everyone does their parts while hiding the complexity.
Inheritance	You create a class called Crew where a crew can be an Actor, Director, StageProducer and all of them has roles, scenes, and filmingSchedules. But they are all still from Crew. Crew is the parent class while Actor, Director, StageProducers are the child ones which inherited some properties like roles, scenes, and filmingSchedule.
Polymorphism	Imagine you have a class Crew. Then you shouted “1, 2, 3! Action!!”. The director will watch them, the videographers will film them and the actors will begin to do what they are supposed to do. When you said “action” they do act. But their actions are different and unique to each other. And regardless, they are still a part of Crew.

2. Virtual Science Laboratory (10 points)

Scenario: A school is developing a Virtual Science Laboratory where students can conduct experiments, use laboratory equipment, and observe chemical or physical reactions.

Principle	Your Example
Encapsulation	Imagine you created a class called Chemical. Then Chemical has properties like viscosity, odor, texture. Now, those properties are hidden and is set on a private variable instead of a public variable so that just in case, the information within the chemicals are hidden and protected. And it can only be altered or changed using a specific method that has the ability to do so such as setProperties().
Abstraction	You have a function called measure(). Now after calling that function it returns to you the value of the measurements you are looking for. However, it did not show you the complexity or the process underneath it. You got the temperature, and changeInWeight. But behind the scenes, multiple attempts and measurement in temperature happened then the mean or average of it was calculated then you got the measurement.
Inheritance	Imagine you created a class called Chemical. Then Chemical has properties like viscosity, odor, texture. Then there are these MuriaticAcid, BakingSoda, HydrogenPeroxide. They are all a chemical where the Chemical is the parent class then those mentioned are the child class. They share the same trait. They have viscosity, odor, texture.
Polymorphism	Let's say you create a class called MeasuringTool which has a property or the ability to measure. Then you have different kinds of measuring tools like beaker, spoon, thermometer, barometer, ruler, measuringTape. They all have the ability to measure except thermometer measures temperature, beaker measures liquid, and ruler measures the length of something. But they are still a measuring tool.

3. Mystery Detective Agency (10 points)

Scenario: A detective agency is developing a Mystery Case Management System that manages detectives, witnesses, suspects, evidence, and investigations.

Principle	Your Example
Encapsulation	Let's say that you have a class called Person. Now that person has evidence. In order to protect that evidence, it must be hidden and only accessible to specific methods within that class. So, to do that, you set the evidence as private that can only be changed and accessed with the function editEvidence().
Abstraction	Imagine that you have a function called searchEvidence(). After calling that function, it gave you the evidences that you are looking for. But underneath it is a very long process where the detective searched for clues then the investigator interrogated some person. After that the data are correlated and then fact checked. And only after verification, you received valid evidence. You only used the function searchEvidence() and it gave you data. Even though inside it is a very complicated process.
Inheritance	Let's use the class Person again who has an age, name, and most of all is the evidence. Now, imagine that each person is different. Someone is a detective, a witness, a suspect, or an investigator. They all share the same trait because they are all a person even though they are different. Meaning to say that detective, suspect and others inherited the traits from Person.
Polymorphism	Now think of an action. Suppose the function searchEvidence() again. The witness recalled their memory, then the detectives looked for clues and then the investigator interrogated some person. They all searched for evidence but it has many forms. The searching of evidence was unique and distinct for each person.